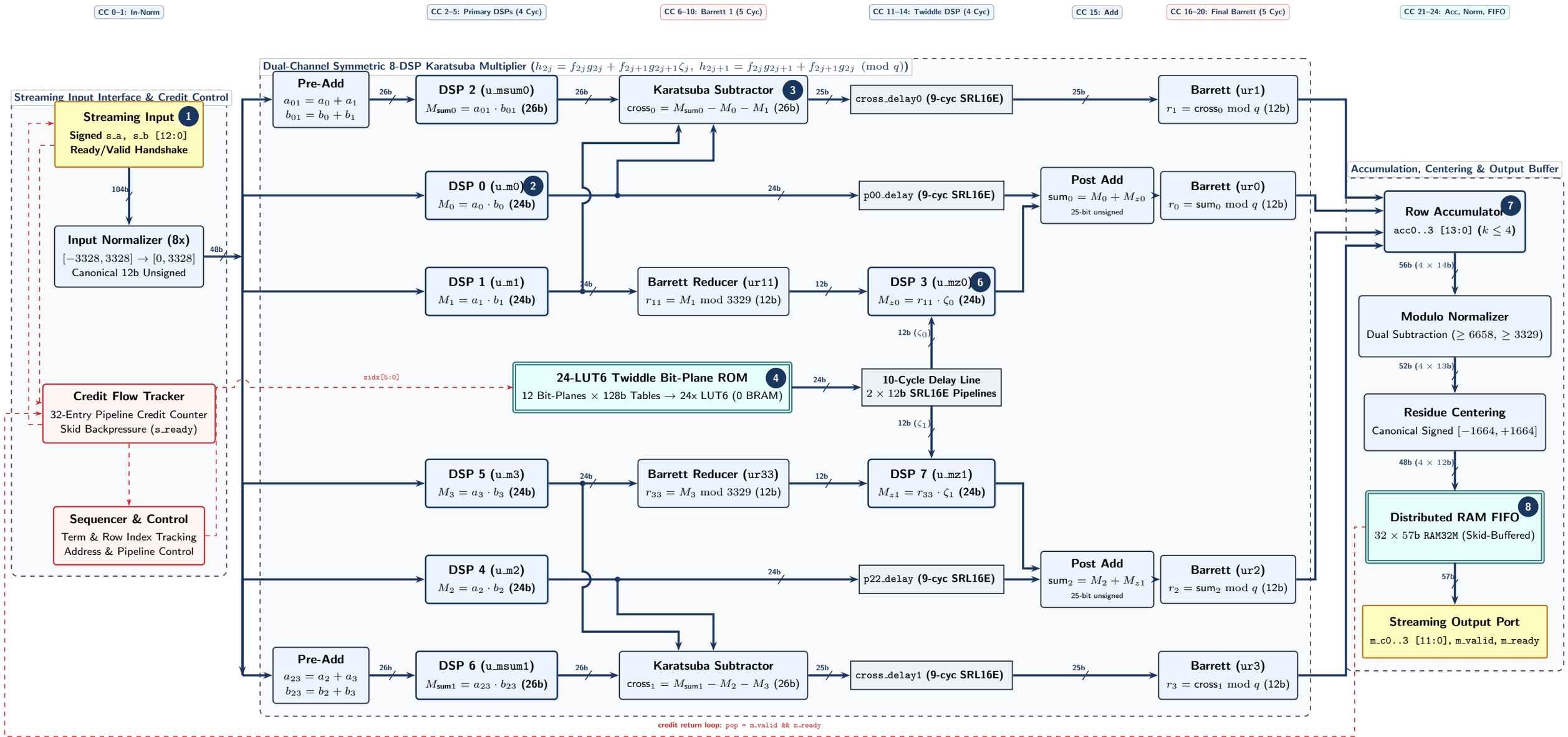
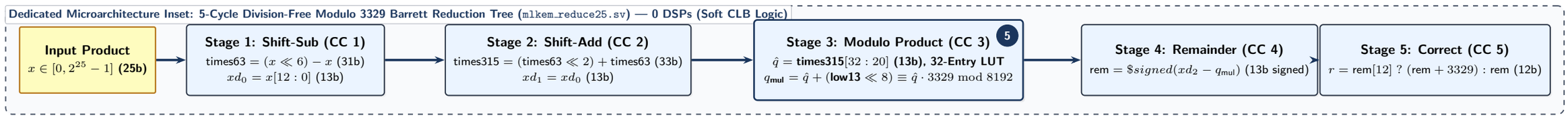
ML-KEM HPKA: Dual-Channel 8-DSP Karatsuba Pointwise Multiplier

Subsystem 06 Microarchitecture — Streaming Arithmetic & Modular Reduction Datapath (`mlkem_mul_stream.sv` & `mlkem_reduce25.sv`)





ID	Component / Signal	Width / Format	Description
1	Streaming Input & Credit Control	8 × 13b In, 6b Credit	Converts signed $[-3328, 3328]$ inputs to canonical $[0, 3328]$; tracks 32-entry pipeline credits to assert `s_ready`.
2	Primary Multiplier DSP Bank	6 × DSP48E2 Slices	Dual 3-slice banks evaluating $a_0b_0, a_1b_1, (a_0 + a_1)(b_0 + b_1)$ with 4 internal pipeline registers at 450 MHz.
3	Karatsuba Exact Integer Subtractor	26-bit Soft Carry Logic	Single-cycle subtractor evaluating cross terms $\text{cross} = M_{\text{sum}} - M_0 - M_1$, eliminating 2 DSP slices per pair.
4	Twiddle Bit-Plane ROM & Pipeline	24x LUT6 Logic (0 BRAM)	Evaluates 12 bit-planes of 128-bit truth tables mapped to 24 LUT6 primitives with 10-cycle SRL delay matching.
5	Modulo 3329 Barrett Reduction Tree	25-bit to 12-bit Soft CLB	5-cycle division-free Barrett reduction exploiting $q = 13 \times 2^8 + 1$ with $M = 315$ and 32-entry LUT, consuming 0 DSPs.
6	Twiddle Multiplier DSP Slices	2 × DSP48E2 Slices	Multiplies quadratic products by aligned twiddles: $M_{z0} = r_{11}\zeta_0$ (DSP 3) and $M_{z1} = r_{33}\zeta_1$ (DSP 7).
7	Row Accumulator & Centering Unit	14-bit / 12-bit Soft CLB	Accumulates 4 coefficient residues for $k \in \{2, 3, 4\}$, applies dual modulo subtraction, and centers to $[-1664, 1664]$.
8	Elastic Distributed RAM FIFO	32 × 57b RAM32M	32-entry distributed LUTRAM buffer storing 4 centered coefficients and tags with skid-buffered output flow control.